Beta-Oxidation of Fatty Acids

Definition

- **Beta-Oxidation may be defined as the oxidation of fatty acids on the β-carbon atom.**
- **This results in the sequential removal of a two carbon fragment, acetyl CoA.**

Stages and tissues

- Three stages
- Activation of fatty acids - in the cytosol
- Transport of fatty acids into mitochondria
- Beta-Oxidation proper in the mitochondrial matrix
- Fatty acids are oxidized by most of the tissues in the body.
- Brain, erythrocytes & adrenal medulla cannot utilize fatty acids for energy requirement

Fatty acid activation

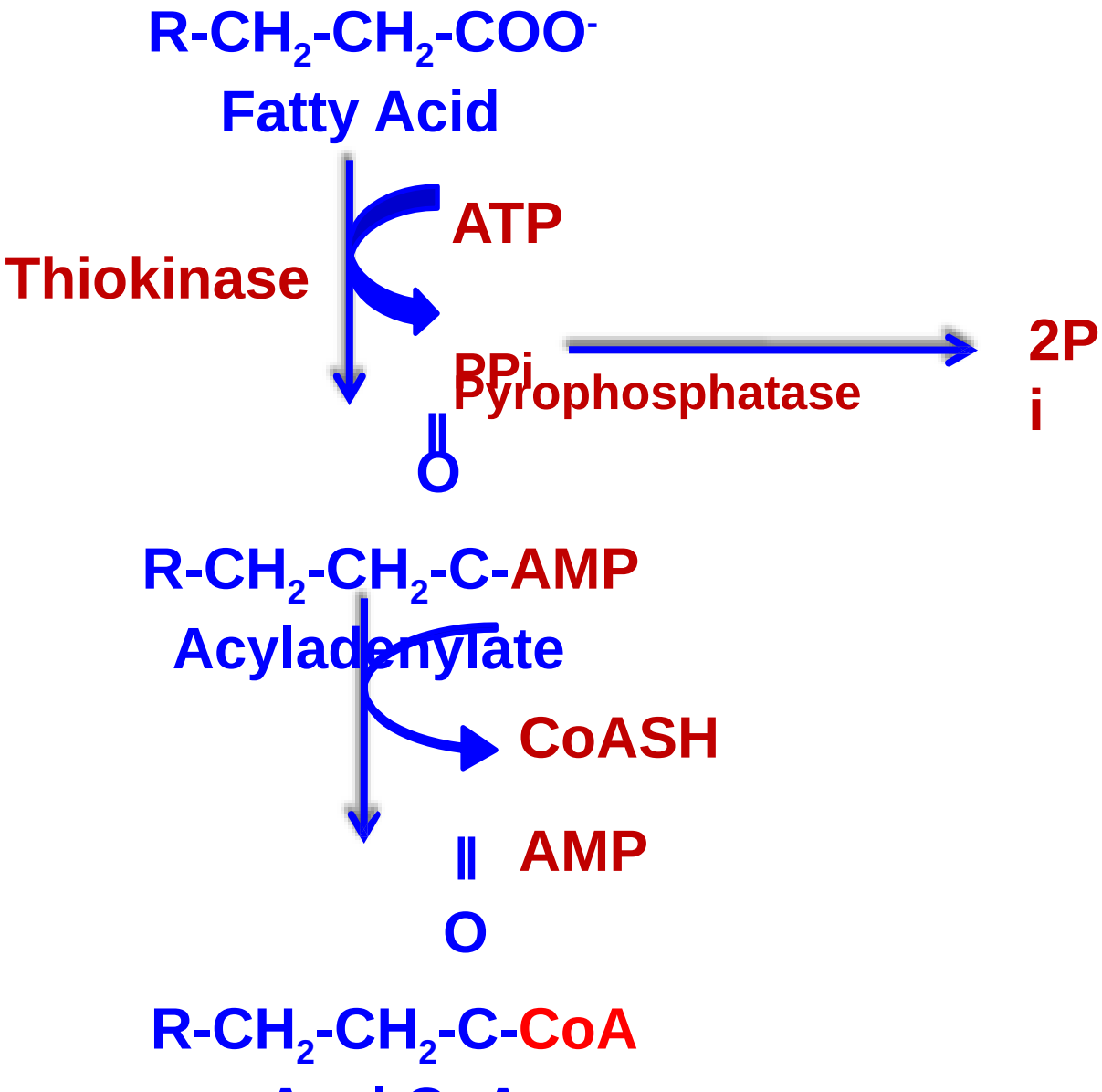
- Fatty acid activation taking place in cytoplasm.
- Fatty acids are activated to acyl CoA by thiokinases or acyl CoA synthetases.
- The reaction occurs in two steps & requires ATP, coenzyme A and Mg^{2+}
- Fatty acid reacts with ATP to form acyladenylate which then combines

- Two high energy phosphates are utilized, since ATP is converted to pyrophosphate (PPi).
- The enzyme inorganic pyrophosphatase hydrolyses PPi to phosphate.
- The immediate elimination of PPi makes this reaction totally irreversible.

Thiokinases

- **Three different enzymes, one each for short chain, medium chain & long chain fatty acids.**
- **Small chain fatty acids may also be activated by thiophorase enzyme, using succinyl CoA.**

Activation of fatty acid to Acyl CoA



Transport of Acyl CoA into Mitochondria

- The inner mitochondrial membrane is impermeable to fatty acids.
- A specialized carnitine carrier system (carnitine shuttle) operates to transport activated fatty acids from cytosol to the mitochondria.
- Carnitine is β-hydroxy γ-trimethyl aminobutyrate, synthesized by lysine & methionine in liver & kidney.

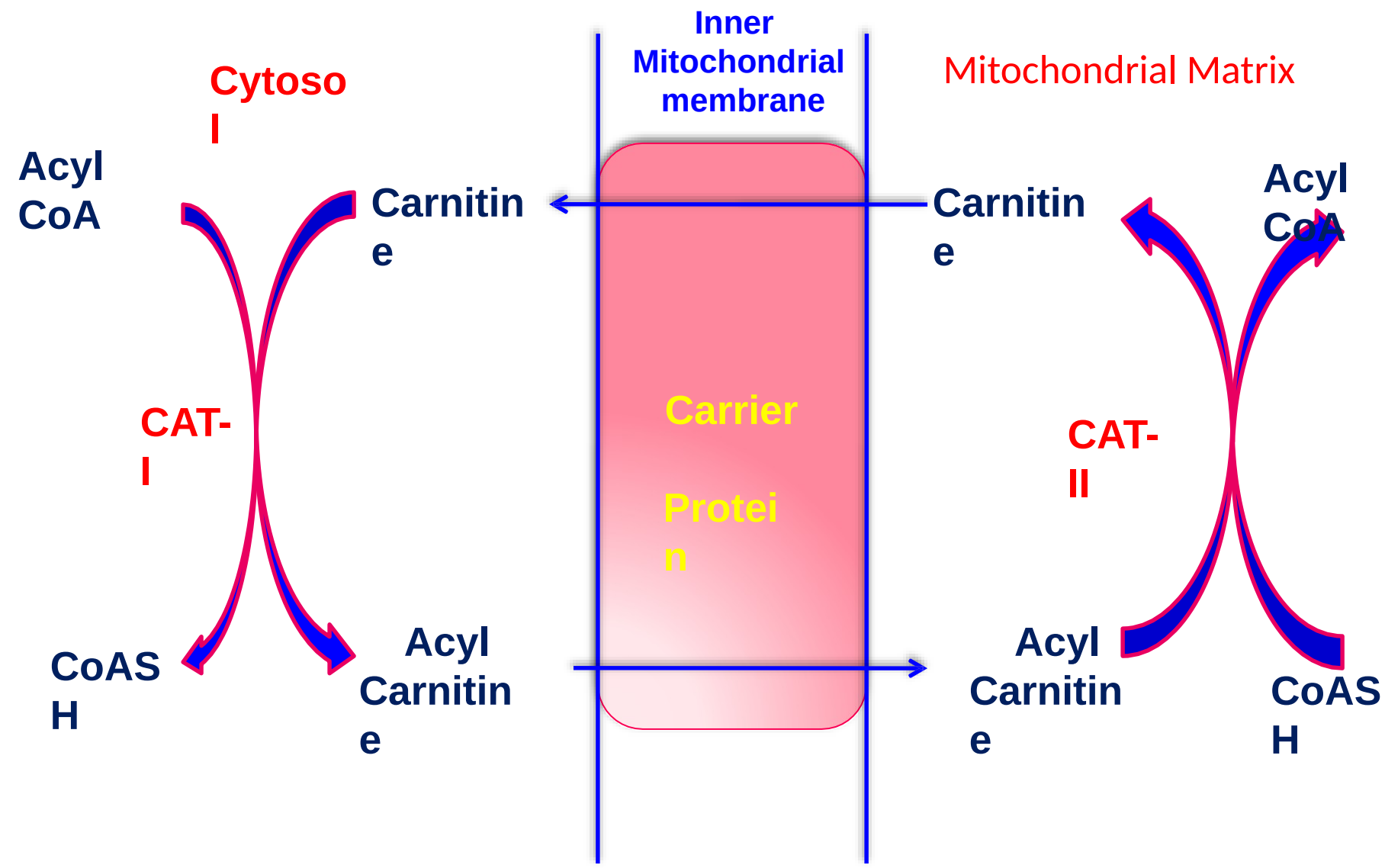
Stages

- It occur in four stages.

1. Acyl group of acyl CoA is transferred to carnitine catalyzed by carnitine acyltransferase I (CAT-I) (present on the outer surface of inner mitochondrial membrane).
2. The acyl-carnitine is transported across the membrane to mitochondrial matrix by a specific carrier protein (Translocase).

3. **Carnitine acyltransferase II (CAT-II)** (found on the inner surface of inner mitochondrial membrane) converts acyl-carnitine to acyl CoA.
4. The carnitine released returns to cytosol for reuse by translocase.

Carnitine transport system



β-Oxidation Proper

- **Each cycle of β –oxidation, liberating a two carbon unit-acetyl CoA, occurs in a sequence of four reactions**

1. **Oxidation**
2. **Hydration**
3. **Oxidation**
4. **Cleavage**

1. Oxidation

- Acyl CoA undergoes dehydrogenation by an FAD-dependent flavoenzyme, acyl CoA dehydrogenase.
- A double bond is formed between α & β carbons (i.e., 2 & 3 carbons)

2. Hydration:

- Enoyl CoA hydratase brings about the hydration of the double bond to form β - hydroxyacyl CoA.

3. Oxidation

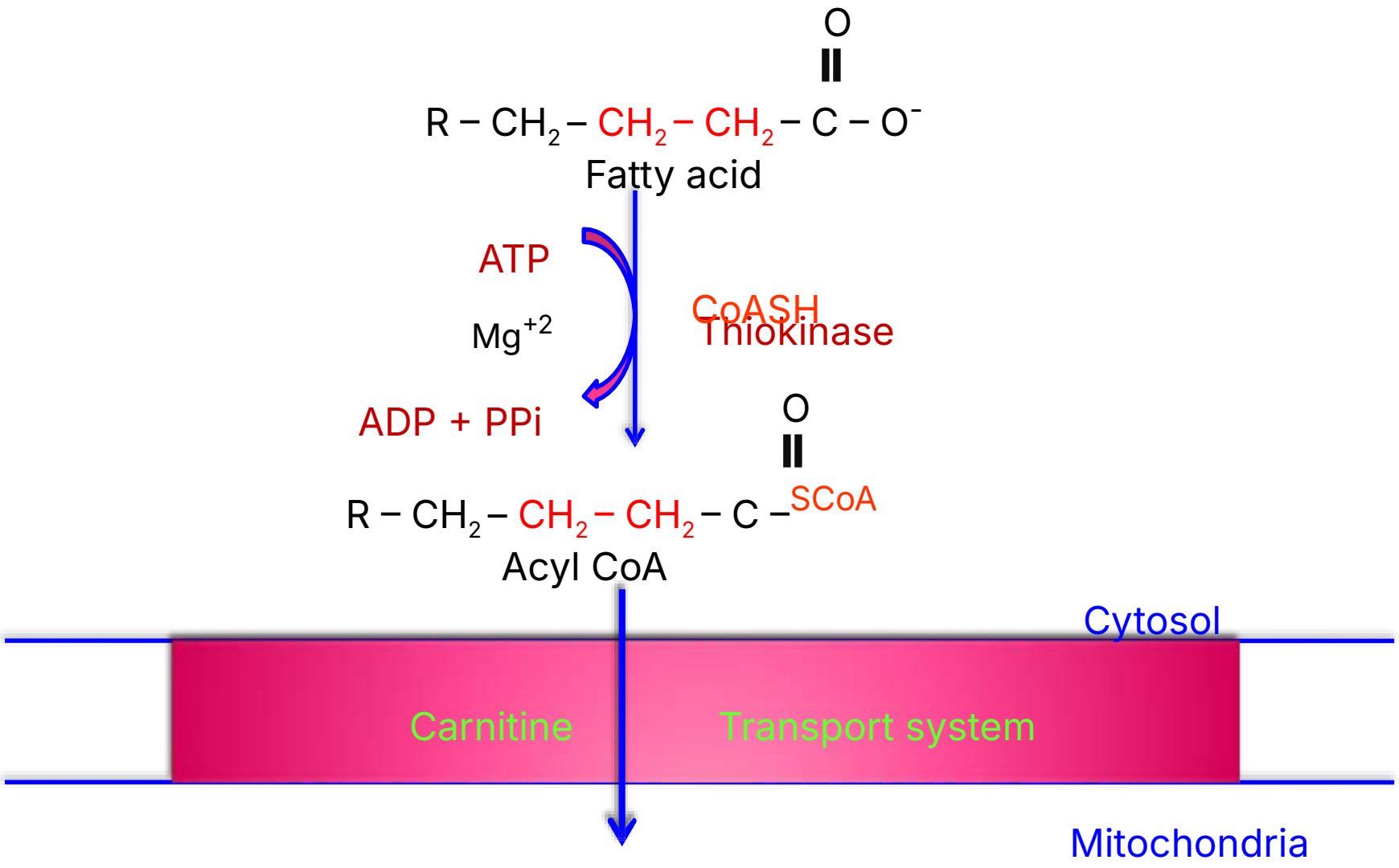
- β-Hydroxyacyl CoA dehydrogenase
  - catalyses the second oxidation & generates NADH.
- The product formed is β-ketoacyl CoA.

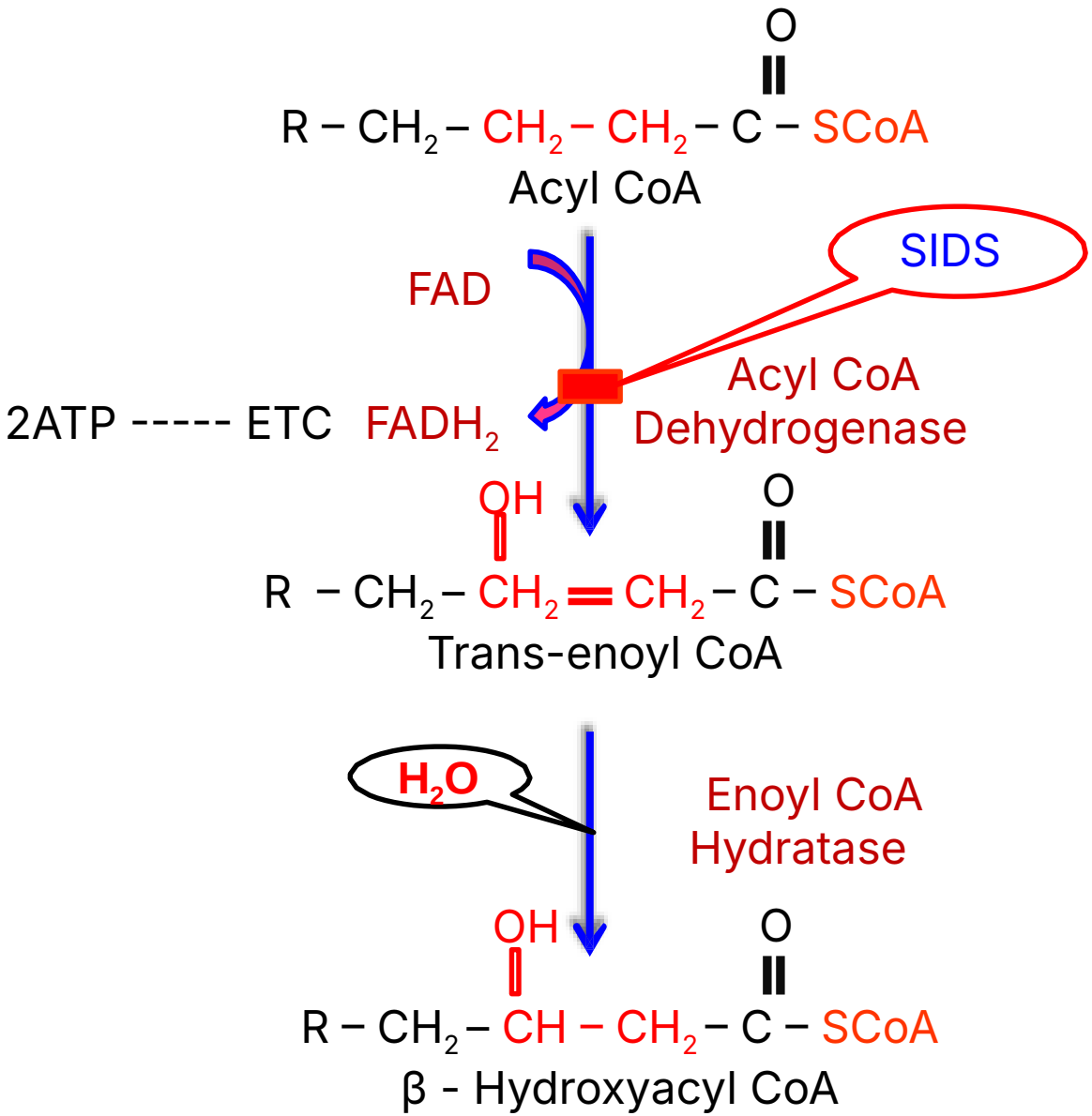
•4. Cleavage

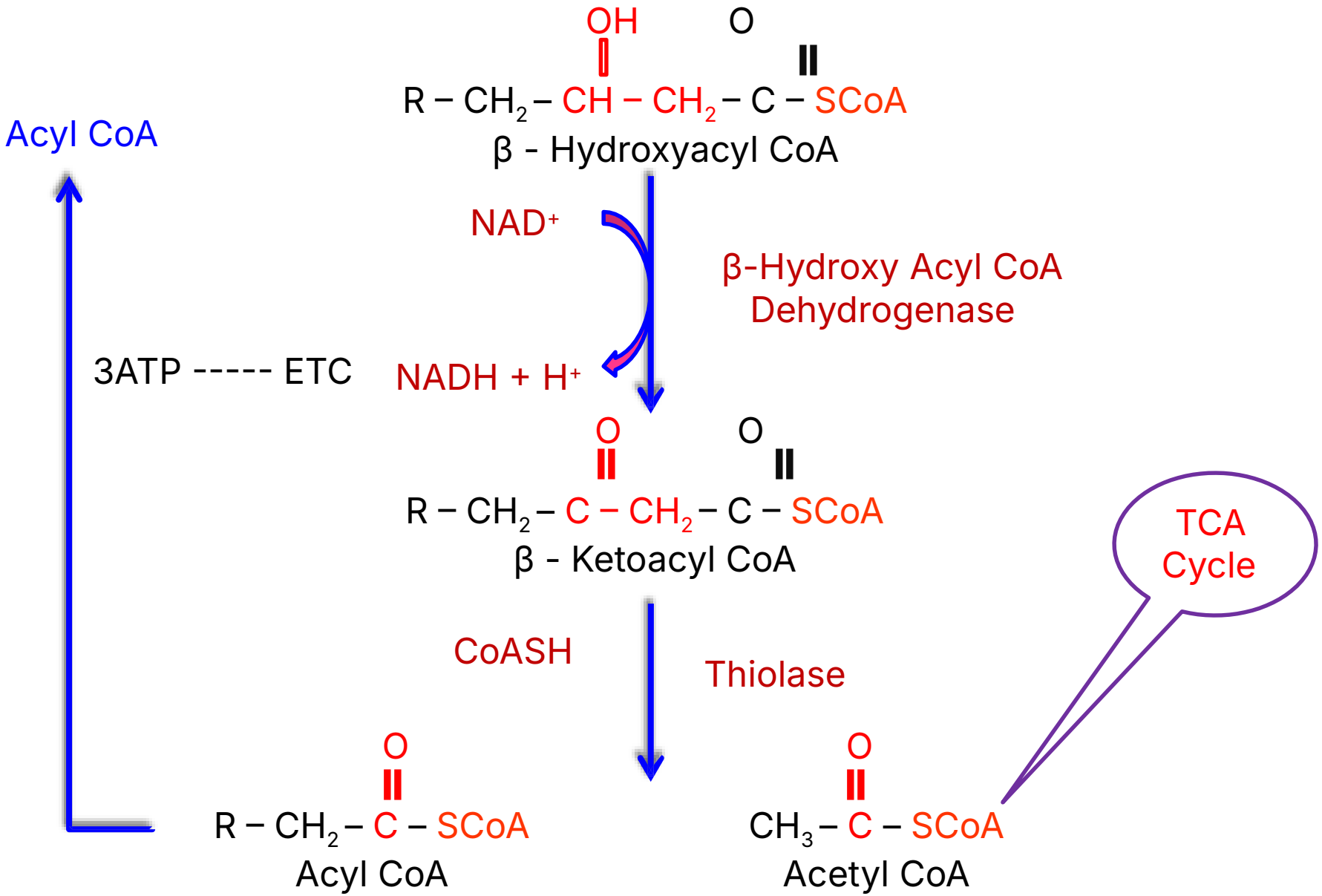
- The final reaction in β -oxidation is the liberation of a 2 carbon fragment, acetyl CoA from acyl CoA.
- This occurs by a thiolytic cleavage catalysed by β- ketoacyl CoA thiolase (or thiolase).

- **The new acyl CoA, containing two carbons less than the original, reenters the β- oxidation cycle.**
- **The process continues till the fatty acid is completely oxidized.**

β-Oxidation of fatty acids




O
R – CH2 – CH2 – CH2 – C – SCoA
Acyl CoA
FAD
SIDS
Acyl CoA
Dehydrogenase
2ATP ----- ETC
FADH2
OH
O
R – CH2 – CH2 = CH2 – C – SCoA
Trans-enoyl CoA
H2O
Enoyl CoA
Hydratase
OH
O
R – CH2 – CH – CH2 – C – SCoA
β - Hydroxyacyl CoA


OH
O
R – CH2 – CH – CH2 – C – SCoA
β - Hydroxyacyl CoA
Acyl CoA
NAD+
β-Hydroxy Acyl CoA
Dehydrogenase
3ATP ----- ETC
NADH + H+
O
O
R – CH2 – C – CH2 – C – SCoA
β - Ketoacyl CoA
TCA
Cycle
CoASH
Thiolase
O
O
R – CH2 – C – SCoA
Acyl CoA
CH3 – C – SCoA
Acetyl CoA

Oxidation of palmitoyl CoA

- Palmitoyl CoA + 7 CoASH + 7 FAD^+ + 7 NAD^+ + 7 H_2O → 8 Acetyl CoA + 7 $FADH_2$ + 7 NADH + 7H^+
- Palmitoyl CoA undergoes 7 cycles of β - oxidation to yield 8 acetyl CoA.
- Acetyl CoA can enter citric acid cycle & get completely oxidized to CO_2 & H_2O.

Energetics of β -oxidation

Mechanism	ATP yield
I. β- 0xidation 7 cycles 7 FADH2 [Oxidized by Electron Transport Chain (ETC) each FADH2 gives 2 ATP] 7 NADH (Oxidized by ETC, each NADH Liberate 3ATP)	14 21
II. From 8 Acetyl CoA Oxidized by citric acid cycle, each acetyl CoA provides 12 ATP	96
Total energy from one molecule of palmitoyl CoA Energy utilized for activation (Formation of palmitoyl Co A)	131 -2
Net yield of oxidation of one molecule of palmitate	**=129**

Regulation of β-oxidation

- The availability of free fatty acid (FFA) regulates the net utilisation through β- oxidation.
- The level of FFA is controlled by glucagon:insulin ratio.
- Glucagon increases FFA level & insulin has the opposite effect.

- CAT-I is the regulator of entry of fatty acid into mitochondria.
- Malonyl CoA inhibits CAT-I activity.
- Thus during de novo synthesis of fatty acid, the beta oxidation is inhibited.

Sudden infant death syndrome (SIDS)

- **Unexpected death of healthy infants, usually overnight**
- **Due to deficiency of medium chain acyl CoA dehydrogenase.**
- **Glucose is the principal source of energy, soon after eating or feeding babies.**

- **After a few hours, the glucose level & its utilization decrease & the rate of fatty acid oxidation must simultaneously increase to meet the energy needs.**
- **The sudden death in infants is due to a blockade in β -oxidation caused by a deficiency in medium chain acyl CoA dehydrogenase (MCAD)**

Jamaican vomiting sickness

- Characterized by severe hypoglycemia, vomiting, convulsions, coma & death.
- It is caused by eating unriped ackee fruit- contains an unusual toxic amino acid, hypoglycin A.
- This inhibits the enzyme acyl CoA dehydrogenase & β -oxidation of fatty acids is blocked, leading to various complications

- Abnormalities in transport of fatty acids into mitochondria & defects in oxidation leads to deficient energy production by oxidation of long chain fatty acids.
- Features:
- Hypoglycemia, hyperammonemia, weakness & liver diseases.
- Acyl carnitine accumulates when the transferases or translocase is deficient.
- Dietary supplementation of carnitine improve the condition.

Oxidation of odd chain fatty acids

- **Oxidation of odd chain fatty acids is similar to that of even chain fatty acids.**
- **At the end 3 carbon unit, propionyl CoA is produced.**
- **Propionyl CoA is converted into succinyl CoA.**
- **Succinyl CoA is an intermediate in TCA cycle**
- **Propionyl CoA is gluconeogenic**

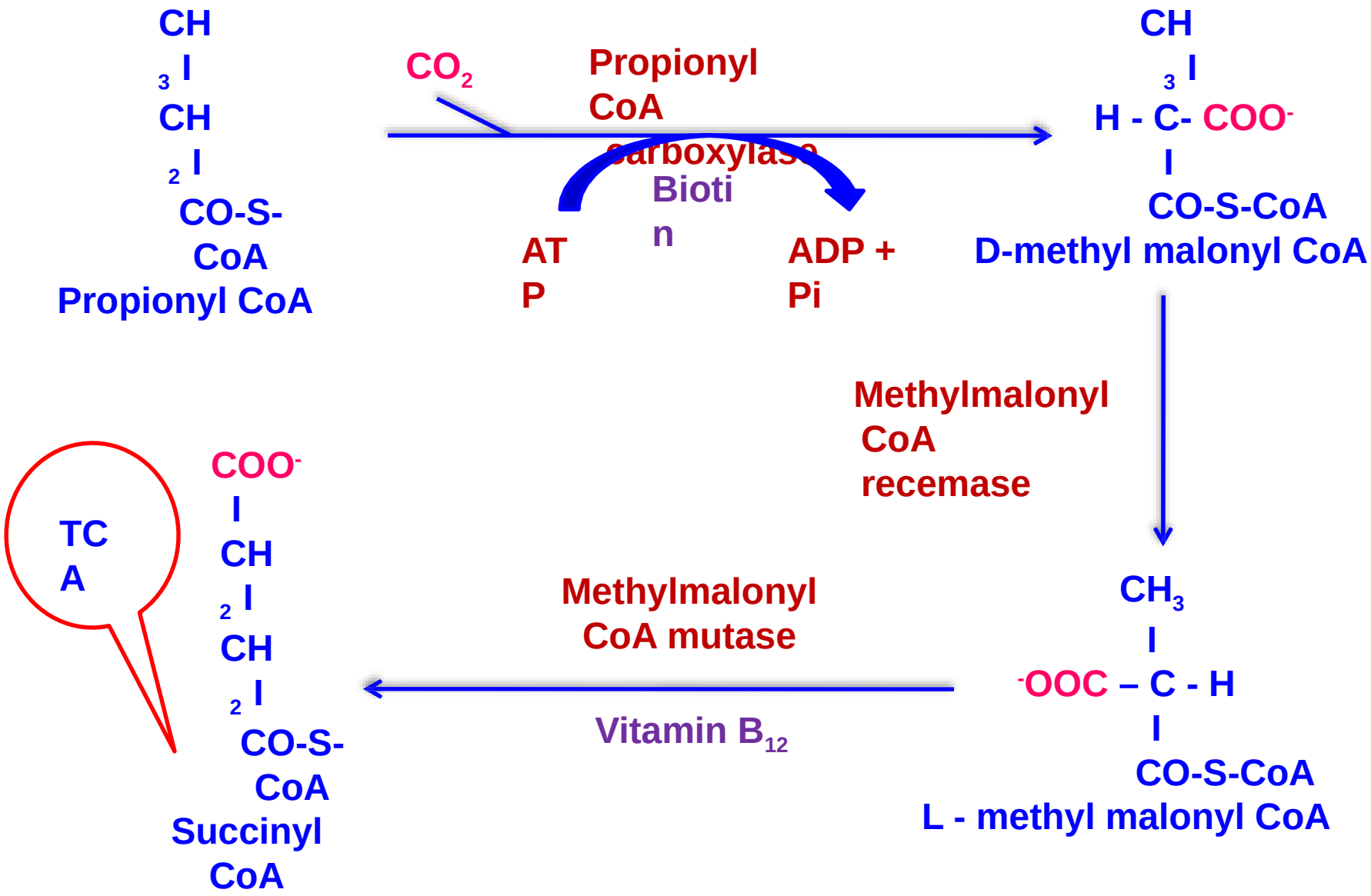
Conversion of propionyl CoA to succinyl CoA

- Propionyl CoA is carboxylated to D-methyl malonyl CoA by a biotin dependent carboxylase.
- Biotin & ATP is utilized in this step.
- Methylmelonyl CoA Recemase:
- Recemase acts upon D-methyl malonyl CoA to give L-methyl malonyl CoA.
  - This reaction is essential for the entry of this

compound into metabolic reactions of body

- **Methylmalonyl CoA Mutase:**
- **Mutase catalyzes the conversion of L-methyl malonyl CoA (a branched chain compound) to succinyl CoA (a straight chain compound).**
- **Mutase is an vitamin B_{12} dependent enzyme.**
- **Succinyl CoA enters the TCA cycle, & converted into oxaloacetate, it is used for gluconeogenesis.**
- **Propionyl CoA is also derived from metabolism of valine &**

Conversion of succinyl CoA to propionyl CoA



Inborn errors of propionate metabolism

- **Propionyl CoA carboxylase deficiency:**
- **Characterized by propionic acidemia, ketoacidosis & developmental abnormalities.**
- **Methyl malonic aciduria:**
- **Two types of methyl malonic acidemias**
- **Due to deficiency of vitamin B_{12}**
- **Due to defect in the enzyme methyl malonyl CoA mutase or recemase.**

- Accumulation of methyl malonic acid in body.
- Methyl malonic acid is excreted into urine.
- Symptoms:
- Severe metabolic acidosis, damages the central nervous system & growth retardation.
- Fatal in early years of life.
- Treatment:
- Some patients respond to treatment with

α-oxidation

- Oxidation of fatty acids on α-carbon atom is known as α-oxidation.
- In this, removal of one carbon unit from the carboxyl end.
- Energy is not produced.
- No need of fatty acid activation & coenzyme A
- Hydroxylation occurs at α-carbon atom.

- It is then oxidized to α-keto acid.
- This, keto acid undergoes decarboxylation, yielding a molecule of CO_2 & FA with one carbon atom less.
- Occurs in endoplasmic reticulum.
- Some FA undergo α - oxidation in peroxisomes.

- α- oxidation is mainly used for fatty acids that have a methyl group at the beta-carbon, which blocks beta- oxidation.
- Major dietary methylated fatty acid is phytanic acid.
- It is derived from phytol present in chlorophyll, milk & animal fats.

Refsum's disease

- **Due to deficiency of the enzyme α-hydroxylase (phytanic acid oxidase)**
- **α – oxidation does not occur.**
- **Phytanic acid does not converted into compound that can be degraded by beta – oxidation.**
- **Phytanic acid accumulates in tissues.**

Symptoms

- Severe neurological symptoms, polyneuropathy, retinitis pigmentosa, nerve deafness & cerebellar ataxia.
- Restricted dietary intake of phytanic acid (including milk-is a good source of phytanic acid)

Omega- oxidation

- Minor pathway, takes place in microsomes.
- Catalyzed by hydroxylase enzymes involving NADPH & cytochrome P-450.
- Methyl (CH_3) group is hydroxylated to CH_2OH & subsequently oxidized with the help of NAD^+ to COOH group to produce dicarboxylic acids.
- When β-oxidation is defective & dicarboxylic acids are excreted in urine causing dicarboxylic aciduria.

Zellweger syndrome

- It is a rare disorder.
- It is characterized by absence of peroxisomes in almost all the tissues.
- Long chain fatty acids are not oxidized.
- Long chain fatty acids are accumulated in tissues-mainly brain, liver & kidney.
- It also known as cerebrohepatorenal syndrome.

References

- Lippincott & Harper
- **Chatterjea**